A Novel Method for Lower Limb Joint Angle Estimation Based on sEMG Signal

Pengjie Qin and Xin Shi

***Abstract*— Due to the strong dynamic and time-varying characteristics of the surface electromyography (sEMG) signal and the coupling between lower limb muscles, it is challenging to accurately estimate the joint angle of complex lower limb motion. Firstly, we proposed a wavelet packet decomposition correlation dimension (WPD_CD) analysis method. A three-level wavelet packet decomposes the sEMG signal, and the optimal and stable feature vector is extracted by the correlation dimension analysis. A singular spectrum nonlinear autoregressive exogenous model (NARX_SSA) is proposed to map the optimal eigenvector to the lower limb joint angle to improve the angle accuracy and smoothness. The experimental results show that this method only needs to collect three lower limb muscles and accurately estimate four movements of knee and hip angles. The overall RMSE was 0.32, the minimum RMSE was 0.20, and the standard deviation (SD) was 20.23. Compared with other methods, the accuracy of angle estimation was improved by six times, and the smoothness was improved by 13%. Therefore, this method can be effectively applied to the angle estimation of complex lower limb motion patterns.**

***Index Terms*— Angle estimation, correlation dimension analysis, nonlinear autoregressive model, singular spectrum smoothing, surface electromyography (sEMG), wavelet packet decomposition.**

I. Introduction

SURFACE electromyography (sEMG) signals are related to the human body's expected action [1]–[3]. The sEMG has the advantages of high resolution, mature acquisition technology, and noninvasive acquisition [4]. It is the most commonly used bioelectrical signal in modern man–machine interface technology. It is often used as a control command to realize the motion control of auxiliary equipment [5]–[8]. Therefore, predicting the human body's continuous joint angle from the sEMG signal has become the key. It is crucial for the motion control of dynamic prosthesis, exoskeleton, and rehabilitation robot.

The motion estimation method based on the sEMG signal mainly uses biomechanical and neural network regression models to establish the relationship between the sEMG signal and human motion. The biomechanical modeling method starts from the musculoskeletal model and establishes the relationship between the sEMG signal and the amount of joint motion [9], [10]. However, the model involves many physiological parameters, making it very difficult to extend the model's application [11].

Compared with establishing a complex physiological model, the mapping relationship between sEMG and exercise based on neural network regression is a more direct and convenient method [12]–[16]. Wang *et al.* [17] proposed a general regression neural network (GA-GRNN) based on a genetic algorithm (GA-GRNN), which used sEMG signal to establish a knee joint angle prediction model. That method can solve the problem that the estimation result is unstable due to the randomness of artificial selection parameters. Deng *et al.* [18] proposed a continuous motion estimation algorithm based on the principal component analysis (PCA) and regularized extreme learning machine (RLEM). The root mean square error was 8.478, and the correlation coefficient was 0.954. Ma *et al.* [19] extracted the root mean square (rms) characteristics of the sEMG signal and used BP neural network to predict the knee joint angle in real time to realize the automatic gait switching of lower limbs. Wang *et al.* [20] proposed a feature extraction method of facial electrical signals based on wavelet packet decomposition and established a prediction model of facial electrical signals based on an extreme learning machine (ELM). The prediction accuracy was 96.23%. Tang *et al.* [21] proposed a sliding window feature extraction method based on the first-order differential signal's energy value and then used BP neural network to establish the estimation model. The results showed that the estimation error of the knee joint angle was 3.25%–11.65%. Li *et al.* [22] proposed a knee-bending moment prediction model based on the BP network and RBF network, and the correlation coefficient between the predicted value and the real knee moment value was above 0.80.

However, there are some problems in the angle estimation of continuous human joints. Firstly, to improve the accuracy of angle estimation, some researchers usually used the sEMG signal of all channels as the input of the neural network model. Other researchers used time domain, frequency domain, or time–frequency domain methods, such as rms, MAV, MPF, wavelet transform, empirical mode decomposition, and other methods, to preprocess the raw sEMG signal and extract the feature vector as the input of the neural network model [23]–[25]. However, with the intense noise and randomness of sEMG signals [26]–[28], there are redundant signals in the raw sEMG signals collected, and

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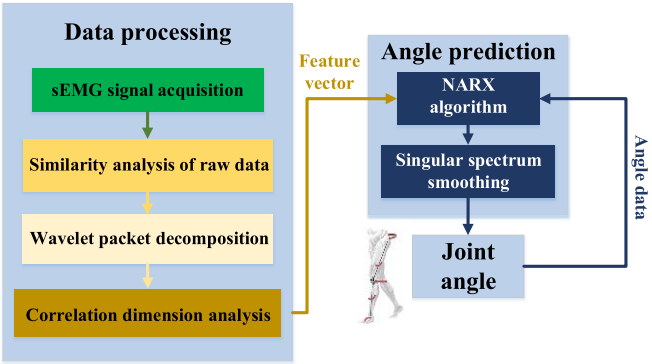

Fig. 1. Entire frame predicts the lower limb joint angle.

the extracted feature vectors find it difficult to describe the motion of lower limb joints accurately. Therefore, how to extract the effective feature vector of the sEMG signal needs further research. Secondly, the neural network model's training time is extended, there is no standard method to define the appropriate network structure, and the performance depends on the input training data. Furthermore, the sEMG signal and joint angle are typical non-stationary, time-series signals, so the trained model is challenging to adapt to the new test data, which will lead to considerable error and fluctuation of angle estimation [29], [30]. However, the traditional neural network model does not use smoothing technology and output feedback to improve angle estimation accuracy and smoothness.

Aiming to solve the above two problems, we proposed a lower limb angle estimation method based on the wavelet packet decomposition correlation dimension (WPD_CD) analysis and the singular spectral smooth nonlinear autoregressive exogenous model (NARX_SSA). Firstly, the raw sEMG signals' similarity was analyzed, and the redundant sEMG signals were eliminated. Then, WPD was used to extract the raw sEMG signal's feature vector. The CD of the feature vector was analyzed, and the optimal and stable feature vector was extracted. The NARX_SSA was proposed to eliminate the local fluctuation of direct mapping. The optimal feature vector was mapped to the knee and hip joint angle to improve angle prediction accuracy and smoothness. The entire framework is shown in Fig. 1.

II. Material and Methods

A. sEMG Signal Acquisition

We used the biometrics DataLITE wireless sEMG system and the W150 wireless angle system to collect sEMG signals and angle signals. The sampling frequency of the system is 500 Hz and the response time of the sensor is less than 3 ms. The transmission delay is less than 1.5 sampling point, which has good real-time performance.

Referring to [28], we collected ten muscles related to lower limb movement: sartorius muscle (SR), medial femoral muscle (VM), rectus femoris (RF), semimembranous muscle (SM), lateral femoral muscle (VL), gracilis (GC), lateral gastrocnemius (LG), biceps femoris (BF), semitendinosus (ST), and medial gastrocnemius (MG), as shown in Fig. 2(a). The lower limb knee joint and hip joint sensors were pasted, as shown in Fig. 2(b).

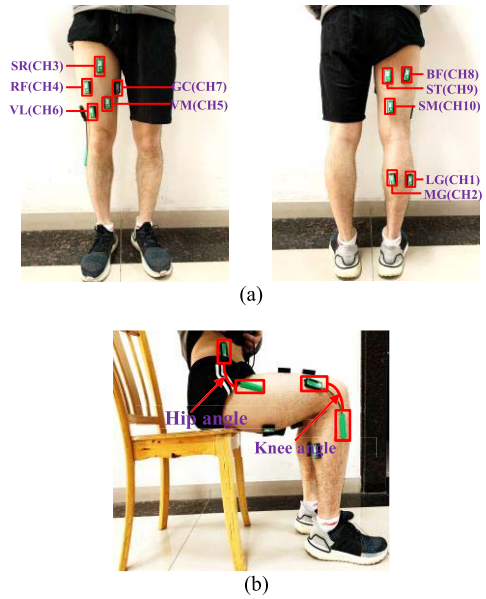

Fig. 2. Sensor paste position. (a) sEMG sensor paste position. (b) Angle sensor paste position.

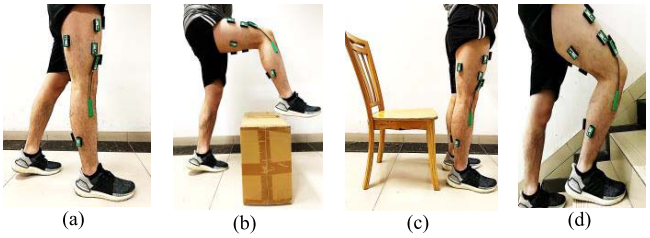

Fig. 3. Four movements of the lower limb. (a) HW. (b) CO. (c) SU. (d) GU.

This study is mainly applied to the application of exoskeletons in logistics handling and military fields, so the research objects selected are young males. We selected five subjects (male, 25 ± 3 years old, body fat rate 17% ± 3%) to participate in the experiment. The lower limb behavior pattern was selected: walking on flat ground (HW), crossing obstacles (CO), standing up (SU), and going up (GU) the stairs, as shown in Fig. 3. All subjects participated in the above four kinds of lower limb movements, and the completion time of each kind of movement was 1.6 s. The procedure followed the declaration of Helsinki.

B. Redundant sEMG Signal Segmentation

We defined X_k as the sEMG signal vector of time k, i.e.,

$$X_k = [x_{1,k}, x_{2,k}, \ldots, x_{10,k}]. \tag{1}$$

As described in the first section, the redundant vector of the sEMG signal will affect the model's accuracy and stability. Therefore, we proposed a cosine similarity method to segment redundant vectors.

First, we calculated the similarity values of different sEMG signal channels as follows:

$$\mathrm{CSV}_{ij} = \cos(\theta) = \frac{\sum_{k=1}^{N} x_{i,k} \times x_{j,k}}{\sqrt{\sum_{k=1}^{N} (x_{i,k})^2} \times \sqrt{\sum_{k=1}^{N} (x_{j,k})^2}} \quad (2)$$

where N is the signal length. CSV_{ij} is the cosine similarity value of muscle activity of channels i and j.

We took CSV_{ij} as an index to judge the redundancy of the signal, and we can divided X_k into two subvectors

$$X_k = \begin{bmatrix} X_k^{(1)} \\ X_k^{(2)} \end{bmatrix} \quad (3)$$

where $X_k^{(1)}$ represents the redundant vector and $X_k^{(2)}$ represents the irredundant vector.

C. Correlation Dimension Analysis Based on Wavelet Package Decomposition

Referring to [31]–[33], we selected the discrete Meyer wavelet to decompose the sEMG signal into three layers, where $\{\mathrm{WPD1}, \mathrm{WPD2}, \ldots, \mathrm{WPD8} | i = 1, 2, \ldots, N\}$ represents the eight decomposition vectors of the wavelet packet.

Since the sEMG signal has strong nonlinearity, randomness, and dynamics [26]–[28], it is easily disturbed by noise. It can be approximated as a chaotic system. Therefore, we proposed a G-P algorithm to analyze the correlation dimension of the wavelet packet decomposition vector and extracted the feature vector, which can accurately reflect the lower limb joints' motion. The main steps are described as follows:

The phase space reconstructed by embedding dimension m and delay time τ is as follows:

$$Y = \begin{cases} Y_1 = (\mathrm{WPD1}_i, \mathrm{WPD1}_{i+t}, \\ \qquad \ldots, \mathrm{WPD1}_{i+(m-1)t}), \quad i = 1, 2, \ldots, N \\ \ldots \\ Y_8 = (\mathrm{WPD8}_i, \mathrm{WPD8}_{i+t}, \\ \qquad \ldots, \mathrm{WPD8}_{i+(m-1)t}), \quad i = 1, 2, \ldots, N \end{cases} \quad (4)$$

where m and τ are determined by the convergence result and the sampling interval, respectively.

Then, the correlation integral is calculated

$$C(r) = \lim_{N \to \infty} \frac{1}{N(N-1)} \sum_{i \neq j} \theta(r - |Y_i - Y_j|) \quad (5)$$

where $|Y_i - Y_j|$ represents the distance between Y_i and Y_j. $C(r)$ is the cumulative distribution function representing the probability that the distance between two points in the phase space is less than r. $\theta(x)$ is the Heaviside function, which satisfies the following conditions:

$$\theta(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0. \end{cases} \quad (6)$$

When N is large enough and r is small enough, the correlation dimension D_m and correlation integral $C(r)$ satisfy the following logarithmic linear relationship:

$$D_m = \lim_{t \to 0} \frac{\ln C(r)}{\ln r}. \quad (7)$$

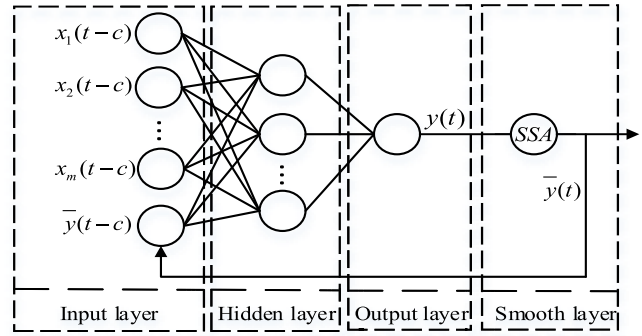

Fig. 4. Angle prediction framework based on NARX_SSA models.

The optimal wavelet packet decomposition vector can be determined through the correlation dimension analysis to reduce noise and redundant information interference.

D. Angle Estimation of the NARX Model Based on SSA

The sEMG signal generated by lower limb joint movement is a complex dynamic and nonlinear process [26]–[30]. Also, the joint angle data are a nonlinear time-series data. Also, the observed angle data are easy to be disturbed by noise. If the sEMG signal is directly mapped to the joint angle, the prediction results will have large fluctuations. Therefore, we proposed an angle estimation method for the NARX_SSA, as shown in Fig. 4, where $x(t)$ represents the eigenvector of the sEMG signal, $y(t)$ represents the angle data, $\overline{y}(t)$ represents the smoothed result, and c is the input and output delay time. Refer [27] and [30] for details of NARX and SSA. Its input includes sEMG signal and joint angle data and has strong dynamic adaptability. It can accurately describe the nonlinear dynamic system and improve the stability of the prediction results. It is suitable for the establishment of a lower limb joint angle prediction model.

E. Performance Evaluation and Statistical Analysis

We used the root mean square error (RMSE) and standard deviation (SD) to evaluate the difference between the predicted results of various models and the true values. The metrics are as follows:

1) *Root Mean Square Error*:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_{\mathrm{true},i} - y_{\mathrm{predict},i})^2}. \quad (8)$$

2) *Standard Deviation*:

$$\mathrm{SD} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (y_{\mathrm{mean}} - y_{\mathrm{predict},i})^2} \quad (9)$$

where N represents the signal length, y_{predict} represents the predicted angle value, y_{mean} represents the sample mean, and y_{true} represents the angle collected.

Besides, three kinds of neural networks, ELMAN and ELM [20], [26], commonly used in estimating lower limb joint angles, are compared with the proposed method.

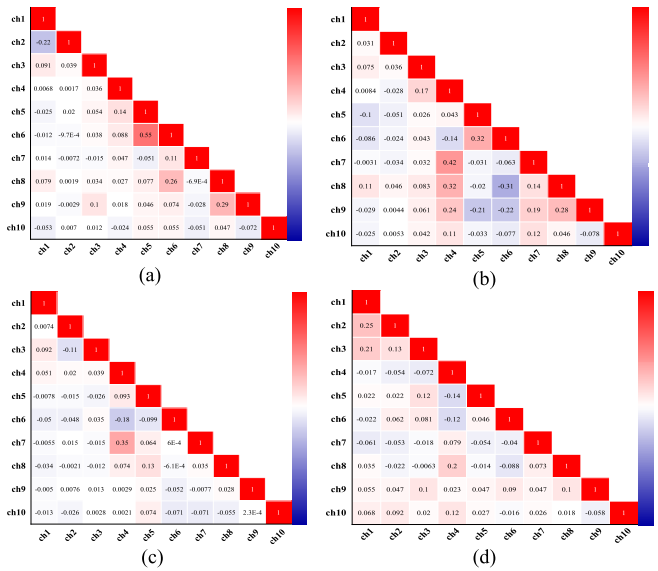

Fig. 5. Similarity coefficient of different motions. (a) Similarity coefficient between the different channels of CO. (b) Similarity coefficient between the different channels of GU. (c) Similarity coefficient between the different channels of HW. (d) Similarity coefficient between the different channels of SU.

III. Experiments and Results

A. Effective sEMG Signal Extraction Based on Similarity Analysis

We calculated the similarity coefficient of all subjects' data. To ensure that the low similarity between each muscle, we set the similarity segmentation threshold to 0.1, as shown in Fig. 5. We divided the sEMG signals of ten muscles into a redundant signal $X_k^{(1)}$ and an irredundant signal $X_k^{(2)}$, as shown in (10)

$$\begin{aligned} X_k^{(1)} &= [x_{4,k}, x_{5,k}, x_{6,k}, x_{7,k}, x_{8,k}, x_{9,k}, x_{10,k}] \\ X_k^{(2)} &= [x_{1,k}, x_{2,k}, x_{3,k}]. \end{aligned} \tag{10}$$

B. Optimal Feature Vector Extraction for Correlation Dimension Analysis of Wavelet Packet Decomposition

We decomposed three raw sEMG signals by a three-level wavelet packet, and calculated the correlation dimension (CD) of each component to extract the optimal feature vector. In the analysis of correlation dimension, the delay time is $\tau = 10$ and the embedding dimension is $m = 10$. The correlation dimension value of the wavelet packet decomposition vector is shown in Fig. 6.

In the correlation dimension analysis, the smaller the correlation dimension is, the more stable the system is and the less susceptible to noise interference. As shown in Fig. 6, the minimum correlation dimension of the three channels' eight wavelet packet components was 0.17. Therefore, we needed to select the wavelet packet component close to the optimal feature vector's minimum correlation dimension. Among them, the correlation dimension values of components CH1_WPD7, CH3_WPD3, and CH8_WPD3 in the four movements are small, but also have differences. It is beneficial to the prediction of the joint angle.

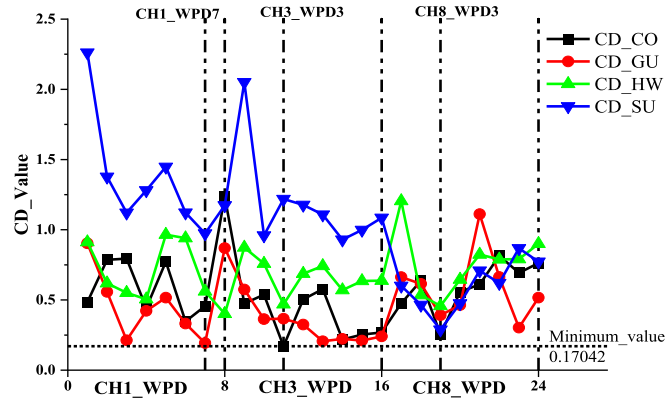

Fig. 6. Wavelet packet decomposition vector correlation dimension numerical graph. CH1_WPD, CH3_WPD, and CH8_WPD represent eight components of the wavelet packet, respectively. CD_Value represents the correlation dimension of each component.

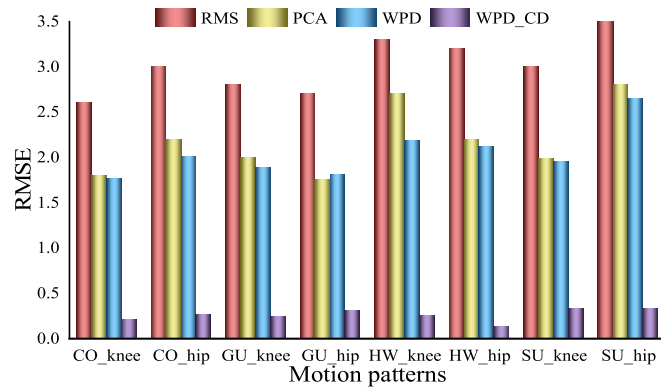

Fig. 7. Average RMSE of hip and knee angle estimation for four movements of lower limbs.

TABLE I

Angle Range of the Knee Joints of Five Subjects

Subjects	CO θ_{knee}^1 / degree	GU θ_{knee}^2 / degree	HW θ_{knee}^3 / degree	SU θ_{knee}^4 / degree
s1	[-1.03,45.27]	[-0.81,30.60]	[-53.41,2.34]	[-1.44,101.8]
s2	[-0.49,42.52]	[-0.22,30.87]	[-66.28,1.12]	[-1.39,91.80]
s3	[-0.81,44.46]	[-0.13,32.58]	[-55.48,3.78]	[-0.81,93.46]
s4	[-0.13,42.79]	[-1.35,35.23]	[-61.50,1.12]	[-1.03,94.05]
s5	[-1.57,46.44]	[-0.67,33.48]	[-55.08,1.75]	[-0.36,99.72]
Mean	[-0.81,44.29]	[-0.63,32.55]	[-58.35,1.57]	[-1.00,96.16]

TABLE II

Angle Range of the Hip Joints of Five Subjects

Subjects	CO θ_{hip}^1 / degree	GU θ_{hip}^2 / degree	HW θ_{hip}^3 / degree	SU θ_{hip}^4 / degree
s1	[-94.0,0.22]	[-68.4,0.85]	[-4.8,24.07]	[-40.2,15.9]
s2	[-86.2,1.93]	[-72.9,0.09]	[-4.0,26.05]	[-40.6,27.0]
s3	[-85.5,0.49]	[-63.9,0.40]	[-2.0,27.13]	[-72.0,31.7]
s4	[-79.5,1.12]	[-67.9,0.67]	[-5.5,28.08]	[-38.6,31.5]
s5	[-81.4,0.85]	[-65.1,1.48]	[-6.0,25.56]	[-41.8,22.0]
Mean	[-85.3,0.92]	[-67.6,0.69]	[-4.4,26.71]	[-46.6,25.6]

C. Model Training

In the process of model training, hyperparameter adjustment has an important influence on the angle prediction results. In the process of NARX_SSA prediction, (8) and (9) were used as evaluation metrics to determine that the number of neurons in the hidden layer was 20 and the delay coefficient

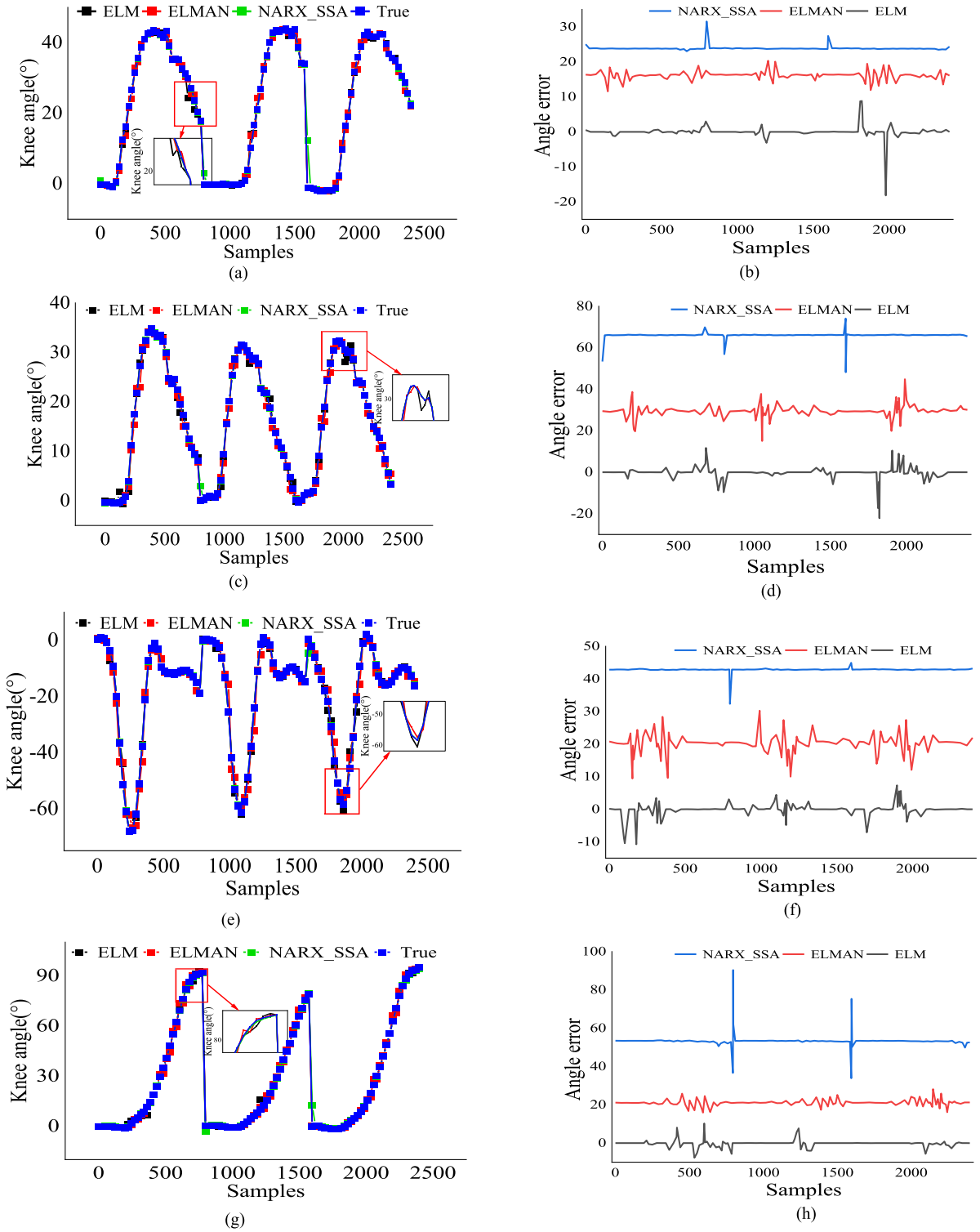

Fig. 8. Comparison of the knee angle estimation in three models. (a) Angle estimation of the knee joint in CO motion. (b) Angle estimation error of the knee joint in CO motion. (c) Angle estimation of the knee joint in GU motion. (d) Angle estimation error of the knee joint in GU motion. (e) Angle estimation of the knee joint in HW motion. (f) Angle estimation error of the knee joint in HW motion. (g) Angle estimation of the knee joint in SU motion. (h) Angle estimation error of the knee joint in SU motion.

was 2. In the ELM prediction process, when the number of hidden layer neurons is 400, the best prediction result can be obtained. In the ELMAN prediction process, the number of training is set to 10 000, the learning rate is 0.01, the training error is 0.001, and the number of hidden neurons is 50, which can achieve better estimation accuracy. To verify the above prediction model, we selected 2400 samples (about 4.8 s) as the test dataset.

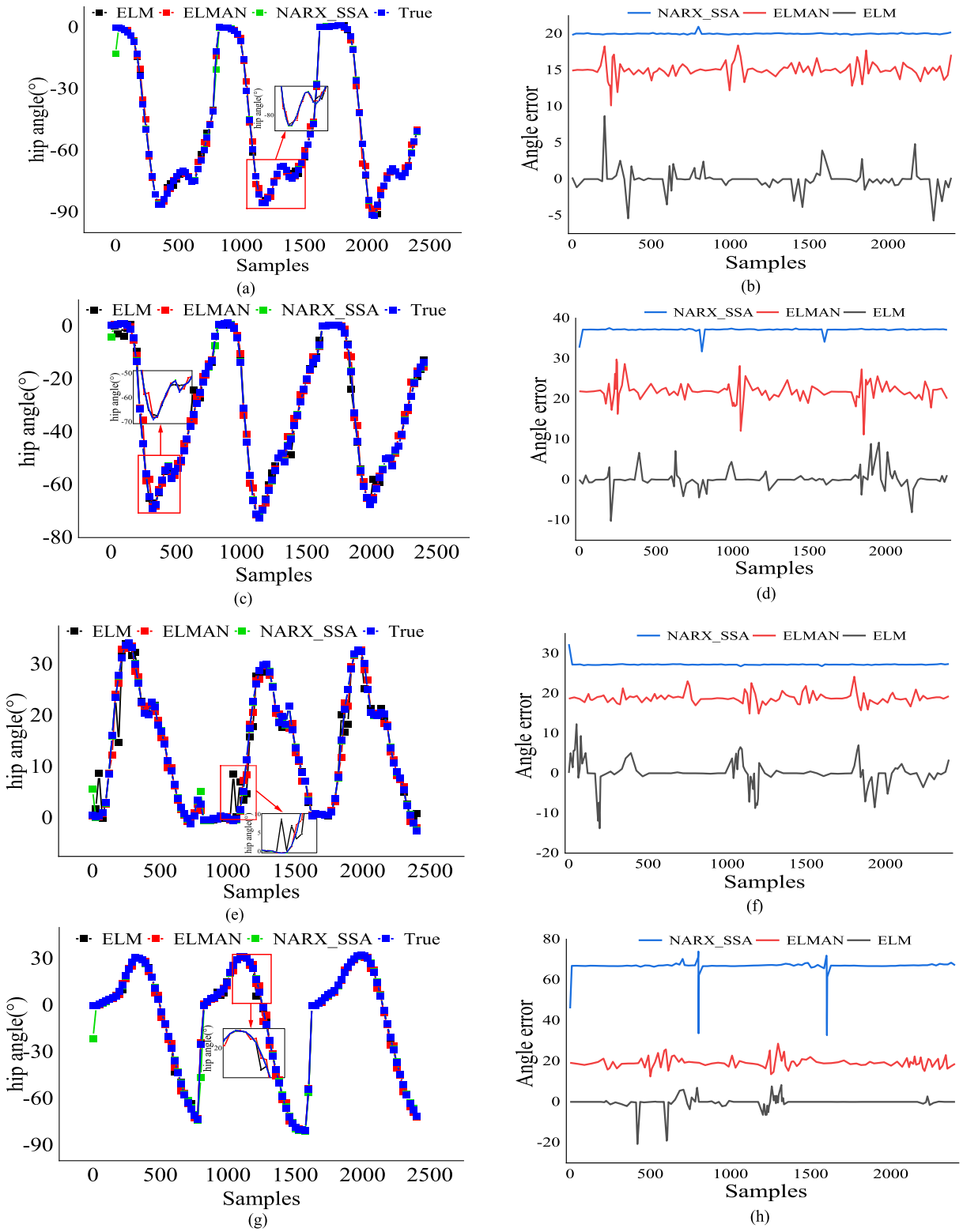

Fig. 9. Comparison of the hip angle estimation in three models. (a) Angle estimation of the hip joint in CO motion. (b) Angle estimation error of the hip joint in CO motion. (c) Angle estimation of the hip joint in GU motion. (d) Angle estimation error of the hip joint in GU motion. (e) Angle estimation of the hip joint in HW motion. (f) Angle estimation error of the hip joint in HW motion. (g) Angle estimation of the hip joint in SU motion. (h) Angle estimation error of the hip joint in SU motion.

D. Results Comparative Analysis

We compared three feature analysis algorithms with the WPD_CD method. They were rms [9], [19], PCA [18], and WPD [20]. We used the NARX_SSA model to estimate the knee and hip angles of the four movements for the lower limbs. The prediction results are shown in Fig. 7.

We can see from Fig. 7 that the average RMSE of the WPD_CD feature extraction method is smaller than the other

TABLE III

FIVE SUBJECTS HIP AND KNEE JOINT ANGLE ESTIMATION RMSE METRICS STATISTICS

Subjects	Algorithms	CO		GU		HW		SU	
		θ^1_{knee} / degree	θ^1_{hip} / degree	θ^2_{knee} / degree	θ^2_{hip} / degree	θ^3_{knee} / degree	θ^3_{hip} / degree	θ^4_{knee} / degree	θ^4_{hip} / degree
	ELM	1.16	1.17	0.91	1.68	1.43	2.57	1.14	1.87
s1	ELMAN	1.19	2.49	1.22	2.47	2.99	1.20	1.50	2.14
	NARX_SSA	0.21	0.33	0.06	0.17	0.18	0.09	0.58	0.84
	ELM	1.46	1.16	1.01	1.25	1.21	2.48	1.03	1.71
s2	ELMAN	1.09	2.44	1.13	2.15	2.09	1.26	1.41	2.62
	NARX_SSA	0.11	0.38	0.16	0.47	0.15	0.06	0.38	0.82
	ELM	1.23	1.02	1.45	1.57	1.56	2.16	1.22	1.11
s3	ELMAN	1.39	2.42	1.13	2.45	2.35	1.14	1.16	1.29
	NARX_SSA	0.31	0.28	0.41	0.24	0.23	0.12	0.18	0.71
	ELM	1.08	1.34	2.05	1.66	1.57	1.96	1.35	1.56
s4	ELMAN	1.67	1.88	1.83	2.55	1.15	2.18	1.56	1.34
	NARX_SSA	0.26	0.16	0.24	0.34	0.31	0.18	0.27	0.67
	ELM	0.98	1.15	1.75	1.46	1.53	1.87	1.43	1.46
s5	ELMAN	1.37	1.41	1.78	2.47	2.03	1.63	2.26	2.13
	NARX_SSA	0.12	0.14	0.34	0.29	0.40	0.21	0.26	0.69
	ELM	**1.18**	**1.17**	**1.43**	**1.54**	**1.46**	**2.25**	**1.23**	**1.54**
Mean	ELMAN	**1.34**	**2.13**	**1.41**	**2.42**	**2.12**	**1.49**	**1.58**	**1.90**
	NARX_SSA	**0.20**	**0.26**	**0.24**	**0.30**	**0.25**	**0.13**	**0.33**	**0.32**

TABLE IV

STATISTICAL ANALYSIS OF STANDARD DEVIATION OF HIP JOINT AND KNEE JOINT IN FIVE SUBJECTS WITH FOUR LOWER LIMB MOVEMENTS

Algorithms	Subjects					Mean
	s1	s2	s3	s4	s5	
ELM	22.93	22.75	22.74	23.01	23.42	**22.97**
ELMAN	23.96	23.44	23.15	23.56	23.87	**23.59**
NARX_SSA	20.77	19.97	20.45	19.65	20.33	**20.23**

TABLE V

MUSCLE SELECTION WITH DIFFERENT SEGMENTATION THRESHOLDS

Segmentation threshold	$X_k^{(1)}$	$X_k^{(2)}$
0.1	$[x_{4,k}, x_{5,k}, x_{6,k}, x_{7,k}, x_{8,k}, x_{9,k}, x_{10,k}]$	$[x_{1,k}, x_{2,k}, x_{3,k}]$
0.2	$[x_{2,k}, x_{3,k}, x_{6,k}, x_{7,k}, x_{8,k}, x_{9,k}]$	$[x_{1,k}, x_{4,k}, x_{5,k}, x_{7,k}]$
0.3	$[x_{6,k}, x_{7,k}, x_{8,k}, x_{9,k}]$	$[x_{1,k}, x_{2,k}, x_{3,k}, x_{4,k}, x_{5,k}, x_{10,k}]$

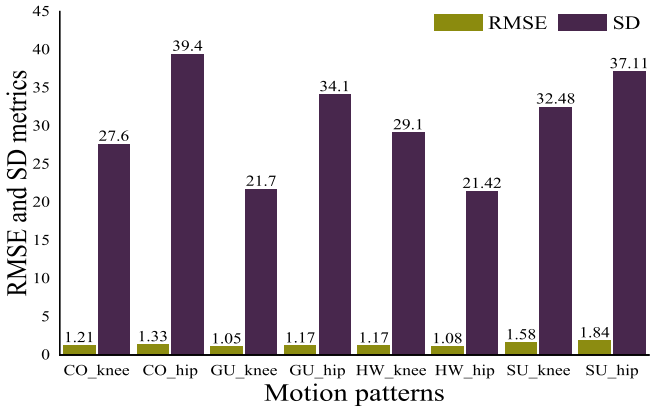

Fig. 10. RMSE and SD of the NARX_SSA model for hip and knee joint.

three methods. The average RMSE of the WPD method is close to the PCA method.

Tables I and II list the range of hip and knee angles for all subjects' four movements. In the lower limb movement process, the angle range of the knee joint and hip joint of CO and GU is similar. Only HW and SU have a great difference in the range of knee angles. Therefore, it is necessary to predict the lower limbs' joint angles more accurately and distinguish different behavior patterns.

To verify the effectiveness of the proposed algorithm, we mainly compared it with the common angle estimation algorithms ELM and ELMAN. As shown in the partially enlarged figure and error diagram in Fig. 8, the prediction results of ELM and ELMAN models were generally close to the true values, but there are still large prediction errors. Compared with ELM and ELMAN prediction results, the NARX_SSA model is closer to the true value of knee joint angle estimation, and the prediction results are smoother, which is conducive to the stability of control. Especially in the movement process of GU and SU, ELM, and ELMAN prediction results have great volatility, NARX_SSA prediction results can well track the true value, as shown in the partially enlarged drawings in Fig. 8(c) and (g).

As shown, when the prediction results and the error diagram in Fig. 9 is compared with ELM and ELMAN, the NARX_SSA model has a smaller estimation deviation of hip joint angle and smaller fluctuation caused by noise. As shown in the partially enlarged figure in Fig. 9(e), in HW motion, ELM and ELMAN predicted values fluctuate greatly, while the accuracy and smoothness of the NARX_SSA model are higher.

We also calculated the RMSE of the hip and knee joint angle estimation by (8), as shown in Table III. The RMSE of the three models is small, ranging from 0.20 to 2.42. The RMSE of NARX_SSA is between 0.20 and 0.74. Compared with ELM and ELMAN models, the estimation accuracy was improved more than six times. The results showed that this method has a good angle estimation effect.

TABLE VI
PREDICTION METHOD OF LOWER LIMB JOINT ANGLE PROPOSED IN THIS ARTICLE WAS COMPARED WITH OTHER METHODS

Author	Number of sensors	Sensors	Motion patterns	Methods	RMSE	References
Feng Zhang et al.	7	FlexComp sEMG sensor and InclinoTrac angle sensor	2	*M*th order nonlinear model and BPNN	7.75	[10]
Yanxia Deng et al.	6	Trigno sEMG sensor and Coda Motion 3D angle system	1	PCA and extreme learning machine (ELM)	8.47	[18]
Lipeng Wang et al.	5	Trigno wireless EMG sensor and Codamotion angle sensor	3	Wavelet packet transform (WPT) and ELM	3.45	[20]
Guoyuan Tang et al.	4	Self-designed portable sEMG sensor and JY901 angle sensor	3	Energy value of the differential signal (DSWE) and BPNN	6.51	[21]
Junhong Wang et al.	5	DELSYS Trigno Wireless sEMG sensor and Codamotion angle sensor	1	Wavelet transform (WT) and ELMAN network	4.16	[26]
Ziyou Li et al.	12	Trigno Wireless sEMG sensor and Vicon 3D angle system	2	Temporally smoothed and MLP	7.71	[28]
Propose method	**3**	**Biometrics DataLITE wireless sEMG sensor and W150 wireless angle sensor**	**4**	**Cosine similarity, WPT, Correlation dimension, and NARX_SSA**	**0.32**	**This work**

Additionally, we used (9) to calculate the standard deviation (SD) of the predicted angle to evaluate the angle estimation's smoothness. As shown in Table IV, the SD of angle estimation of the NARX_SSA model is 20.23, and the smoothness was improved by 13% compared with ELM and ELMAN models. It showed that the angle data predicted by this method has low dispersion. Therefore, the angle estimation result of the NARX_SSA model is smoother, which could improve the stability of robot control.

We selected a new female subject, 165 cm tall and 22 years old. Then, we collected 800 samples as test samples. It can be seen from Fig. 10 that the RMSE and SD of angle estimation are close to those in Tables III and IV, which proved that the method has good adaptability.

We also analyzed the influence of the segmentation threshold on the experimental results. As can be seen from Fig. 5, the positive correlation values range from 0 to 0.5. Therefore, we manually set the segmentation thresholds to 0.1, 0.2, and 0.3. As shown in Table V, different segmentation thresholds can select different muscles. We used the NARX_SSA method to estimate the angle of $X_k^{(2)}$ vector corresponding to each threshold in Table V, as shown in Fig. 11.

As can be seen from Fig. 11, the RMSE of the CO_knee, CO_hip, GU_knee, HW_knee, and HW_hip reaches the minimum when the segmentation threshold is 0.2. Nevertheless, the overall RMSE distribution reaches the minimum when the segmentation threshold is 0.1, and the angle estimation result is more stable.

Moreover, we compared other angle estimation methods, as shown in Table VI. In these studies, most researchers used more sensors to estimate the motion angle, but they considered fewer motion patterns, which is difficult to meet daily activities' needs. However, the algorithm only needs to calculate three channels of the sEMG signals to estimate the four motion angles of the lower limb accurately, and the accuracy of angle estimation is significantly higher than the existing methods.

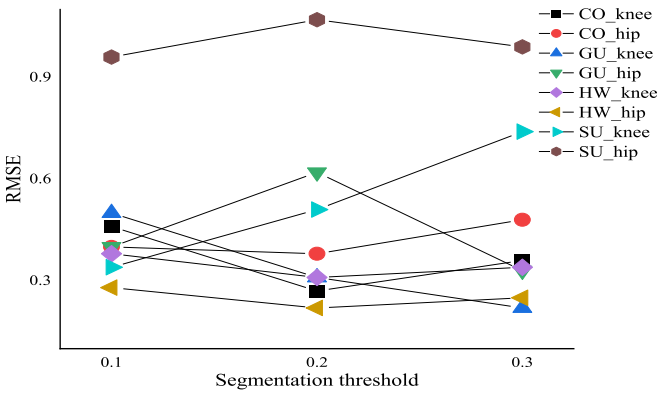

Fig. 11. Angle estimation RMSE under different segmentation thresholds.

Few sensors can make users feel comfortable in practical applications, reduce redundant information calculation, and improve angle estimation accuracy. In addition, the sEMG signal sensor and the angle sensor come from the same system, and the data have good temporal and spatial consistencies, which is conducive to accurate angle estimation. The research results have significant application advantages.

IV. CONCLUSION

We proposed a method to estimate a singular spectrum nonlinear autoregressive exogenous model's lower limb angle based on wavelet packet decomposition and correlation dimension analysis. It only needs to collect sEMG signals of three lower limb muscles to estimate the four motion patterns of the knee and hip angles accurately. We used the cosine similarity method to remove redundant sEMG signals and used WPD_CD to extract the optimal feature vector to eliminate redundant feature vector interference. Then, the singular spectrum nonlinear autoregressive exogenous model (NARX_SSA) proposed in this paper was used to predict lower limb motion's knee and hip angles. The results show that the overall RMSE

was 0.32, the minimum RMSE was 0.20, and the SD was 20.23. The test results of the newly selected subjects also achieved good results. Moreover, the accuracy and smoothness of the NARX_SSA model are better than ELM and ELMAN models, the accuracy of angle estimation was improved by six times, and the smoothness was improved by 13%. The novel method can be effectively applied to the angle prediction of complex lower limb movement mode, and help to improve the motion control performance of sEMG signal device. Also, this method can improve the application effect of exoskeleton motion control in logistics handling and military fields.

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